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Older adults' self-perception, technology anxiety, and intention to use digital public services ³

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Abstract ⁶

To improve the happiness of the older adults enjoying digital public services, this study examines the structural relationship among self-perception of aging, subjective well-being, technology anxiety, self-efficacy, perceived usefulness, and intention to use digital public services for older adults in the context of digital public services. We employ Structural Equation Modeling (SEM) for empirical analysis ($N=345$, February to October 2023). The negative self-perception of aging may lead to the negative emotions of older adults on technology, which will reduce subjective well-being and increase technology anxiety. Technology anxiety indirectly affects the intention to use digital public services, which has an impact on the intention to use through the perceived usefulness and the complete mediating path of self-efficacy and perceived usefulness. Perceived usefulness is the most important factor affecting the behavior intention of older adults to use digital public services. Our study would provide reference to promote older adults' digital literacy and silver happiness and promote the development of the silver digital economy. ⁷

Keywords Technology anxiety, Older adults, Self-perception of aging, Digital public services, Behavior intention; ⁸ Digital age

Introduction ⁹

According to data from the China Internet Network Information Center (CNNIC), as of June 2023, the number of Internet users in China reached 1.079 billion, an increase of 11.09 million compared with December 2022, and the Internet penetration rate reached 76.4%. Chinese Internet users aged 60 and above accounted for 13%, and 16.9%. The proportion of Internet user group aged 40–59 increased from 33.2% in December 2022 to 34.5%, and

the Internet further penetrated into the middle-aged ¹¹ group. Older adults have emerged as a significant driver of the growth in Internet users. From the perspective of the size and composition of the older adults population, the data of the seventh National Census shows that as of November 2020, China's population aged 60 and above is 264.02 million, accounting for 18.70% of the total population, among which the population aged 65 and above is 190.64 million, accounting for 13.50% of the total population. It is expected that by 2035, the number of older adults aged 60 and above in China will exceed 400 million, accounting for more than 30% of the total population, and entering a severely aging society. In terms of age, the elderly group aged 60 and above is the main group of non-Internet users, with the proportion of non-Internet users aged 60 and above accounting for 41.9% of the total number of non-Internet users in China. In terms of reasons for not surfing the Internet, the proportion of

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non-Internet users who do not surf the Internet because they are “too old/too young” is 15.1%. All of the above shows that although the use of the Internet by older people has increased, the adoption of technology among this group is still a big problem.

Updated and upgraded communication technologies have accelerated the process of digitization in all aspects of life. There is no denying that digitization has improved the convenience of people's lives, and older people are willing to use advanced technology to improve their quality of life. The digital development of public services can help older citizens better enjoy their rights as citizens, for example, seniors can check their insurance information and pension online. Although the frequency of use is not high, this is very important for older citizens. As an important topic in the social welfare and security system, social services deserve to be studied and continuously optimized to enhance citizens' social welfare and happiness in life. Digital public services such as online ticket booking, online car rental, online banking, online registration, mobile phone-generated health codes, electronic social security cards, digital libraries and other digital public services are developing rapidly, and public health emergencies have accelerated the digital transformation of public services. The world has recognized the need for digital public services, especially from the sudden global public health emergencies of the previous few years [36]. Digital public services are emerging, but some citizens are being left behind [19], who are unable to enjoy their due rights [27]. These people mainly come from marginalized groups of society including older adults, rural residents, socioeconomically disadvantaged families and the disabled [37, 43]. People who have a low level of technology use or do not have much access to the Internet will find it difficult to enjoy the convenience of digitalized public services, leading to the gradual exclusion from digital public services [27]. It is a vicious circle for this group and they may end up disconnected from society. Older adults are part of this group, and in an increasingly ageing society, their social welfare needs to be safeguarded and deserves to be studied.

Considering the gray digital divide and social aging, this study aims to examine the factors affecting older adults' intention to use digital public services from the perspective of self-perception and technology anxiety, and provide recommendations for aging improvements in public services.

Literature review ⁴

Gray digital divide ⁵

The concept of the digital divide originated in the United States in the 1990s and was first articulated in the report “Falling Through the Net: A Survey of the ‘Have Nots’ in

Rural and Urban America”. It primarily reflects the generation gap resulting from differential access to information technology across different age groups. Research on the digital divide initially focused on “access” to digital technology (the first digital divide) and its “use” (the second digital divide). The gaps in access and use of digital technology led to a third digital divide, namely, the knowledge gap in the digital era, which is particularly pronounced in the adoption of information technology by older adults [12, 18]. The digital divide specifically relevant to older adults is referred to as the gray digital divide.

Scholars have found that social media platforms may play a crucial role in fostering social exclusion among older adults, contributing to the digital divide [41]. Social media has been at the forefront of digitization as the main medium of mass exposure to messages and interpersonal interactions in recent years. Older adults, due to their lack of digital skills, find it difficult to enjoy the social benefits of social media, which has led to a greater digital divide between them and younger users. Yuan proposed a method for analyzing the digital divide among older adults based on text mining, Baidu Index, and principal component analysis [56]. Both studies arrived at the conclusion that the digital divide persists among older adults across different regions and times.

The ongoing existence and deepening of the gray digital divide pose a significant challenge to the benefits older adults can reap in the digital economy. While digital technology continues to advance, societal services are gradually shifting toward digitization, including online shopping, car rentals, digital payments, and telemedicine. However, many older individuals who are not familiar with the latest technological developments face the risk of being left behind. Consequently, digital public services may provide additional social assistance to users, enhancing their convenience in accessing public services. Nevertheless, the digital divide in the use of digitized public services among older adults from different age groups and social demographics, coupled with social exclusion due to a lack of communication with peers, may limit the potential of these tools and ultimately result in social inequality [41]. Especially with the outbreak of the COVID-19 pandemic, people increasingly rely on information and communication technology (ICT) supported digital media, exacerbating these issues. Digital public services, as part of digital media, have also been in the process of digital transformation in recent years. Since digital public services are not used as frequently as other digital medias, and older adults are not the main users of digital information technology, the level of ageing in this area is not high. For these with a slightly lower level of information technology, the digitization of public services may not be able to provide them with convenience and

may even affect their normal life. For example, an online reservation is required in advance to receive protection money, and older persons who do not know how to operate the service will not be able to get the money in time.

Digital technology adoption among older adults ²

Previous studies have confirmed the benefits of older adults using digital technology, including but not limited to society and self-understanding benefits [23] (e.g., increased self-efficacy and self-cognition), interaction benefits [16, 31, 32] (e.g., increased social possibilities and social support), and daily life convenience benefits [15, 22, 42] (e.g., assistance with work, travel, shopping, and financial management). Internet using, especially in the context of services aimed at improving social participation and inclusivity, may ultimately impact the quality of life for older adults. Therefore, a considerable amount of research consistently concludes that computer and internet engagement are beneficial, providing overwhelmingly positive experiences and outcomes for older adults [28].

Given these benefits, many scholars are dedicated to researching the relationship between older adults and the internet, as well as the factors influencing the technological adoption by older adults. The acceptance of the internet by older adults is generally low, influenced by various factors categorized as internal and external. Internal factors can further be divided into individual and psychological factors. In terms of personal factors, many scholars, by comparing older adults with high and low levels of technology adoption, have found that demographic variables such as age, gender [13, 18], education [5, 13, 18, 34, 51], income [5, 18], marital status [51], daily activities [18], and self-assessed health status [5, 34], have varying degrees of impact on their internet use. Sala and other scholars found that individuals aged 65–70, those with higher education, and/or those cohabiting with a spouse/partner are more likely to use information and communication technology [41]. For psychological factors, in the current era of accelerated digitization of public services, older adults face physiological barriers to accepting and adopting digital services due to factors such as age, health status, and economic obstacles. Insufficient digital literacy, lack of experience in use, and inherent weakness in adapting to new things further impede their ability to adopt digital services. For instance, older adults are more likely to adopt e-book reading devices but less likely to choose smartphones [20]. Simultaneously, a lack of cognitive awareness among older adults (lack of interest, perceived usefulness), concerns about potential risks, and worries about security and reliability can reduce their internet usage rates [33]. In the external

environment, high skill barriers to the use of digital services and a lack of social capital support are crucial adoption barriers for older adults [45]. Encouragement from family and friends also proves to be a powerful predictive factor for older adults' internet use [12].

At present, China is in the convergence period of aging and digitization. How to make digital technology better serve the older adults, turn the “gap” into “inclusiveness”, and improve the happiness of the older adults enjoying digital public services is an important social problem to be solved by the country. At present, many studies on the use of digital technology for the older adults focused on the use of smartphones or online social media platforms, and paid less attention to the use of digital public services for the older adults. More studies chose older people's Internet use as an independent or intermediary variable to study the impact of older adults on mental health and well-being. When establishing the digital technology adoption model for the older adults, many researchers directly adopted previous models such as Technology Acceptance Model (TAM), Unified Theory of Acceptance and Use of Technology (UTAUT), etc., without combining the psychological characteristics of the older adults. Many previous studies on the use of technology in the older adults were more qualitative analyses of the current situation of digital use in older adults, less systematic and complete empirical studies. Therefore, this study incorporates the self-perception of aging theory, self-efficacy theory, etc., hoping to comprehensively explain the psychological and behavioral mechanisms of older adults in adopting digital public services. The self-perception of aging theory provides an understanding of how older adults perceive their own aging process and how this perception influences their sense of well-being as well as anxiety. It serves as a fundamental psychological background that affects their willingness to participate in digital public services. The self-efficacy theory focuses on older adults' belief in their own capabilities to use digital technologies.

In the face of the serious aging of Chinese society and the deepening of the silver digital divide, this study will carry out empirical research by establishing a digital public service adoption model for the older adults. Based on the above model combined with self perception of aging theory, self-efficacy theory and TAM, this study starts from the older adults self-perception, combined with the characteristics of the older adults' use digital public service (e.g., technology anxiety) to establish the older adults digital public service adoption model, analyzes the older adults digital public service adoption of psychological factors and mechanism, positive response to the background of digital public service population aging practice reference, to promote the older adults digital literacy and

silver happiness and promote the development of the silver digital economy is of vital significance.

Theoretical basis and research hypotheses

The self-perception of aging theory is a psychological theory, first proposed by the American psychologist Becca Levy, aiming to explain the individuals' cognition and attitude toward their own aging process [25]. Individuals' negative perception of their own aging can affect their behavior and have a negative impact on their quality of life and health status. The negative self-perception of aging may lead to mental health problems and behavioral limitations, resulting in low subjective well-being. Technology is seen as a threat to established norms and patterns of behavior that allows us to adapt to our living environment but bring negative emotional responses, anxiety and fear [30]. Studies have shown that individuals who do not use technology do not change their behavior as they age, and they feel "afraid" and "anxious" about technology [51], and have greater computer-related fear and anxiety than younger adults [1]. There are negative aging stereotypes in society, which individuals accept in social interactions and internalize them into self-perception [25]. If older adults hold a negative self-perception of aging and believe that their physical and cognitive decline affects their learning and use of technology, then they may be less willing to be exposed to new technologies, resulting in technological anxiety (Chen and Persson [5]), . Older people with high subjective well-being tend to receive support from their families, friends or society [6, 24], which may help them reduce technology anxiety, and having support networks can provide resources for older adults to learn and adapt to new technologies. We therefore propose the following hypotheses:

H1: Negative self-perception of aging has a negative impact on their subjective well-being.

H2: The negative self-perception of the older adults has a positive impact on their technology anxiety.

H3: The higher subjective well-being of the older adults has a negative impact on their technology anxiety.

The older adults are affected by their physiological aging, psychological problems will be gradually exposed. Older adults may experience memory loss, decreased attenuation, a slower processing of information, and emotional fluctuations, including low mood, anxiety, loneliness, and depressive symptoms [26]. Body changes with age may increase perceived difficulty and negative attitudes toward technology use and create skepticism about online information [3, 52]. Older adults lacking experience in digital skills may experience barriers to using the Internet, increasing their perceived difficulties,

frustration, and anxiety about using technology and developing skepticism about online information [40], This will seriously affect the use of information technology [51].

TAM is a theoretical model used to explain the behavior of information technology adoption by individuals [7], and it has been widely applied in the research on the willingness to use among various groups of people. The model was originally proposed by Fred Davis in 1986 and has been developed and expanded in subsequent studies. The initial TAM of Fred Davis consists of two core constructs: perceived usefulness and perceived ease of use [8]. The model suggests that individuals' decisions about adopting new technology are mainly dependent on whether they find the technology useful to them and whether they think it is easy to use it. However, there are special circumstances in the context of the elderly using smart products. Many older adults lack a clear understanding of the difficulty of using intelligent products, and there may be some stereotypes and cognitive biases about the difficulty of operating intelligent products. Studies have also shown that perceived ease of use does not have an impact on older people's behavior intention to use it [53]. Therefore, this study did not include perceived ease of use variables based on the TAM. Within the framework of the TAM, most studies have confirmed that there is a significant direct relationship between perceived usefulness and technology adoption [55]. In addition, technology anxiety may lead to negative feelings about technology in the older adults. This sense of restlessness may make it easier for older adults to see technology flaws or problems, affecting their perception of its usefulness and thus the breadth and depth of their use [35].

From the perspective of self-efficacy theory and social learning theory, high self-efficacy individuals are more confident in their abilities and show lower anxiety levels in facing difficult tasks. Therefore, individuals with high self-efficacy should have lower levels of technology anxiety [29]. And they have confidence that they can use the technology effectively, so that they can actively assess the usefulness of the technology. However, older people tend to underestimate their knowledge and ability, lack sufficient confidence in the use of digital technology, and have low self-efficacy for digital technology [51]. Studies have established that technology anxiety, self-efficacy and perceived usefulness are important determinants of behavioral intent to adopt, use, and continue using mFit technology [47]. We therefore propose the following hypotheses:

H4: Technology anxiety in the older adults has a negative effect on their self-efficacy.

H5: Technology anxiety of the older adults has a negative impact on their behavior intention to use digital public services.

H6: Technology anxiety in the older adults has a negative impact on the perceived usefulness of their digital public services.

H7: Self-efficacy in the older adults has a positive impact on the perceived usefulness of their digital public services.

H8: Self-efficacy of the older adults has a positive impact on their behavior intention to use digital public services.

H9: The perceived usefulness of digital public services has a positive impact on their behavior intention to use them.

Following the hypothesis, we build the following model, as shown in Fig. 1.

Method 7

Data collection 8

The survey was conducted using a questionnaire between June and October 2023. In terms of research design, electronic questionnaires were distributed through the Credamo platform, WeChat, and Tencent to reach participants from various regions across China. The total number of questionnaires collected was 360, of which 345 were valid. Following the rigorous recommendations from a power analysis [14], the sample size of this study is sufficient to detect R² values of around 0.10 at a significance level of 5% and a power level of 80%.

The definition of “older adults” varies in the literature. In some countries or regions, the age of 50 may be considered a significant turning point in the human life cycle. Specifically, in certain developing countries, 50 years of age may signify the onset of challenges such as declining quality of life, health problems, or reduced income after retirement, which are typically associated with old age. Therefore, in these regions, individuals aged 50 and above

can be considered as the elderly population. In addition, social media use among individuals over 50 years of age has increased greatly [46], the group aged 60 and above is the main group of non-Internet users. Inspired by previous research with individuals aged 50 and above as older adults [44], this study consider the individuals aged 50 and above as the potential respondents.

During the sampling and recruitment process, purposive sampling was employed to select respondents aged 50 and above, with no history of mental illness, in order to ensure the comprehensiveness and validity of the study, as well as to better explore the relevant issues and seek potential solutions. Prior to the official commencement of the survey, the research team disseminated recruitment information through multiple channels, providing detailed explanations of the survey’s objectives and significance, and inviting eligible individuals to participate. This study was approved by the ethics committee of the hospital. Each participant was fully informed about the details of the survey and completed an informed consent. The questionnaire, as the primary data collection tool, began with the question: “Have you used or do you intend to use any of the following digital applications?” This question offered 11 options, including “online ticket booking, digital travel, online appointment registration and payment, electronic health insurance payment, digital library, online education, electronic payment and transfer, online appointment scheduling, online queuing,” among others, to provide participants with a basic understanding of digital public services.

To ensure the validity and reliability of the data collection tools, the following steps were taken: First, during the questionnaire design phase, a thorough review of relevant literature was conducted, and expert consultations were incorporated to ensure that the questionnaire accurately reflected the research objectives and variable measurement needs. Second, regarding the Credamo platform, it features professional questionnaire design and data collection capabilities, such as preventing

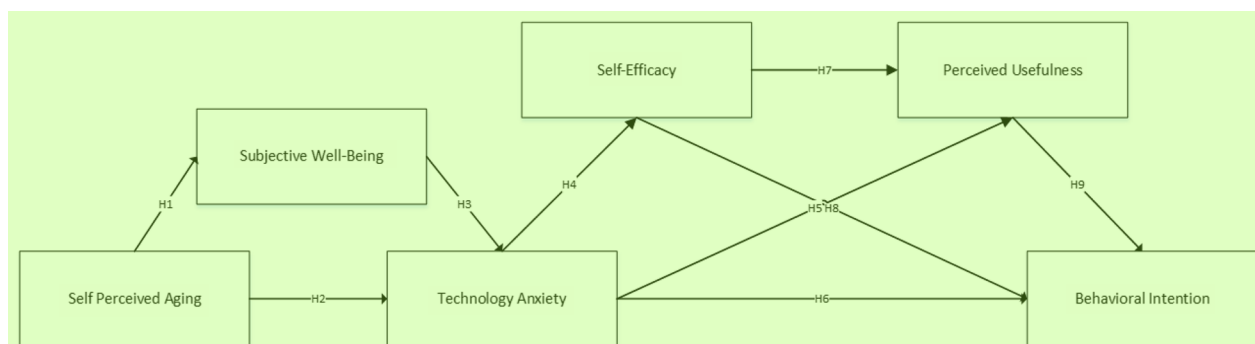


Fig. 1 Model of older adults' intention to use digital public services 15

duplicate responses, intelligently monitoring response times, and offering offline survey services, which help to ensure the authenticity and validity of the data. WeChat and Tencent, as widely used social platforms, enable rapid access to a large pool of potential respondents during the distribution of the questionnaire. Furthermore, by setting specific participation links and conditions, the reliability of the data sources is further ensured. The study primarily designed and published the questionnaire through the Credamo platform, with WeChat and Tencent assisting in the distribution and data collection process. To facilitate respondents in China, the questionnaire items were initially translated from English to Chinese. To ensure translation accuracy, a comprehensive approach combining back-translation, committee review, and pretesting for cross-cultural validation [4] was employed.

Measure 2

Behavior intention (BI) refers to the individual's determination of the subjective probability of taking a specific behavior, which reflects the individual's behavior intention to adopt a specific behavior. We used Venkatesh's subjective norm scale to measure behavior intention [48]. In this study, the Cronbach's α of this scale was 0.845.

Self-efficacy (SE) is the degree of confidence that individuals can complete a specific task or achieve a specific goal. When a person has high self-efficacy, they believe in their ability to meet challenges, solve problems, and learn new skills. Self-efficacy can motivate individuals to try new tasks and persist to achieve success. We used the Huang's subjective norm scale to measure self-efficacy [17]. In this study, the Cronbach's α of this scale was 0.825.

Technology anxiety (TA) expresses concern or fear of the new technology. This concept can be used to explain the difficulties or unease that older adults may have worried about using digital technology. Technology anxiety may reduce their willingness to embrace digital. We used the Venkatesh's subjective norm scale to measure technology anxiety [38, 49, 54]. In this study, the Cronbach's α of this scale was 0.899.

Perceived usefulness (PU) is a core concept in TAM to explain individual perceptions of useful technology. Whether older adults consider digital technologies useful for their daily life will influence their decision to adopt these technologies. We used Davis et al.'s subjective normative scale to measure perceived usefulness [9, 38, 50]. In this study, the Cronbach's α of this scale was 0.847.

Subjective well-being (SWB) is a psychological concept used to describe an individual's subjective evaluation of their quality of life and happiness. We used Diener et al.'s subjective norm scale to measure subjective well-being [11]. In this study, the Cronbach's α of the scale was 0.777,

but the Cronbach's α after the deletion item entitled "If I can start my life again, I will hardly change anything" was $0.813 > 0.777$, thus deleting this item and re-testing the Cronbach's α result was 0.813.

Self-perception of aging (SPA) is the subjective perception of aging. This concept can be used to explain whether older adults believe that digital technology can help them cope better with aging. If they believe that digital technologies can improve their quality of life, they may be more willing to adopt them. We used the Levy et al.'s subjective normative scale to measure the self-perception of aging [25]. In this study, the Cronbach's α of this scale was 0.801.

Except for SPA, all items were measured using the Likert 7-point scale (1=strongly disagree, 2=disagree, 3=slightly disagree, 4=neither agree nor disagree, 5=slightly agree, 6=agree, 7=strongly agree). For measures of SPA, participants answered "yes" or "no" for each item, which scored "1" and "2", respectively. SPA 3 and SPA 5 were scored in the reverse direction so that all items measure negative self-perception of aging. Respondents responded in response to SPA 4 by choosing "better", "worse" or "same". To make SPA 4 comparable with the other four terms, we changed it to a dichotomous variable, combining the "same" response with the "worse" response (in the analysis in the Results section, we verified that combining the "same" response with the "better" response did not change the effect of the independent variable on survival).

The demographic variables included gender (1=female, 2=male), age (1=50–54 years, 2=55–60 years old, 3=61–64 years old, 4=65–70 years old, 5=over 71 years old), education (1=primary school or below, 2=junior high school, 3=High school, 4=University undergraduate, 5=Master, 6=PhD), marital status (1=single, 2=Being married, 3=divorced, 4=widowed), working status (1=full-time, 2=Part-Time, 3=retirement, 4=unemployment), whether the work (major) is information technology related (1=very relevant, 2=Related, 3=unrelated), network age (1=less than 1 year, 2=1–3 years, 3=3–5 years, 4=more than 5 years), Internet use frequency (1=almost daily, 2=at least once a week, 3=Once-monthly, 4=less than once a month), physical condition (1=healthy, 2=In general, 3=poor), whether to live with the children (1=yes, 2=No), permanent residence (1=city, 2=rural areas).

Demographic characteristics of participants 12

Among the respondents, the proportion of men was 54.2% and that of women was 45.8%. Over half of the sample held bachelor's degrees. The age distribution was mainly concentrated in the 55–60 and 61–64 age groups, accounting for 49.8% and 32.5% respectively. The

vast majority were married, with a proportion of 95.9%. Regarding Internet usage, 76.5% of the respondents have been using the internet for more than five years, and over 90% of the respondents use the internet almost daily. In terms of health status, 80.2% reported being healthy. The proportion of permanent urban residents was 88.1%, and over 70% of families had an annual income exceeding 100,000 yuan (See Table 1).

Results 2

In this study, Structural Equation Modeling (SEM) was employed primarily because it allows for the simultaneous assessment of multiple causal relationships and can handle complex interactions between latent variables. SEM is particularly suitable for testing and validating theoretical models, especially when multiple related variables are involved. This study explores the path relationships among several variables, including technology anxiety, self perceived aging, self-efficacy, and behavioral intention. SEM is effective in testing these paths. Two statistical tools including AMOS and SPSS 25 were used for analysis. SPSS 25 was primarily employed for descriptive statistical analysis and reliability testing of the variables to ensure the basic quality and credibility of the data. AMOS was used for common method bias testing, model construction, model fit evaluation, hypothesis testing, and mediation effect analysis. Furthermore, this study focuses on the mediation effect between technology anxiety and behavioral intention, specifically examining whether self-efficacy and perceived usefulness mediate this relationship. The mediation effect was analyzed using Preacher and Hayes (2008) Bootstrapping that does not contain zero confidence interval indicates significant effect.

Reliability and validity tests 4

This study employs Cronbach's Alpha to examine the reliability and dependability of the questionnaire, and utilizes Composite Reliability (CR) and Average Variance Extracted (AVE) to evaluate the convergent validity of each latent variable in the measurement model. As shown in Table 2, the Cronbach's Alpha coefficients of the six latent variables are all greater than 0.8, indicating that the scale has good reliability and dependability. Besides, The Cronbach's Alpha coefficients of each item after deletion are all smaller than the Cronbach's Alpha coefficients corresponding to the variables to which they belong. This indicates that deleting these items will not improve the reliability of the variables. The CR values of the six latent variables range from 0.808 to 0.899 and are greater than 0.7, demonstrating that the measurement model in this study has excellent internal consistency. Moreover, the AVE values of the latent variables are between 0.458

and 0.691 and are all greater than 0.4, suggesting that the convergence and validity of the measurement model are acceptable. The results of the reliability and validity tests are presented in Table 2.

Common deviation test 7

To avoid a common method bias, Harman's univariate method [39] was used. Common method variance test was performed; the number of common factors was set to 1, and the confirmatory factor analysis (CFA) was performed using AMOS. The result fitting index ($\chi^2 / df=6.464$, RMSEA=0.126, SRMA=0.1639, CFI=0.745, TLI=0.721) indicated that all the indicators were far from the critical value. Using the two-factor model method, model A and benchmark model B fit well. Indicators (A-B) show that, $\Delta RMSEA=0.003$, $\Delta SRMA=0.005$, $\Delta CFI = -0.034$, $\Delta TLI = -0.042$. The change of RMSEA and SRMR was less than 0.05, and the change of CFI and TLI was less than 0.1, indicating that the model was not significantly improved after the addition of the common method factor control. Both methods indicated that there was no serious common method variance in the study data and would have no impact on subsequent hypothesis testing.

Model fit test 9

This study uses AMOS software to conduct a model fit test, aiming to assess the goodness of fit between the model and the data. The test typically includes several important fit indices, such as χ^2 / df , RMSEA, CFI, TLI, and IFI. When the model meets these indices, it indicates a good fit. As shown in Table 3, all the fit indices in this study meet the required standards, indicating that the model exhibits a good fit.

Structural equation modeling and hypothesis test 11

In the case of excellent model fitting, we observed the actual situation of each path. As shown in Table 4, the critical ratio of C.R. were all greater than 1.96. In addition to "TA-> BI ($\beta=0.048$, $P=0.456>0.05$)" and "SE-> BI ($\beta=0.096$, $P=0.350>0.05$)", all other pathway significance probability values P were significant ($P<0.05$). The standardized coefficient for path "SPA-> TA", "SE-> PU", and "PU-> BI" was greater than 0 respectively, indicating the above pathways presented a positive correlation. The standardized coefficient for path "SPA-> SWB", "SWB-> TA", "TA-> SE", and "TA-> PU" was less than 0 respectively, indicating these pathways all showed significant negative effects. From the above analysis, H1-H4, H6, H7 and H9 were supported, while H5 and H8 were not supported.

Technology anxiety, self-efficacy, perceived usefulness, and behavioral intention constitute the typical chain

Table 1 Demographic characteristics of participants ¹

Variable	Classification	Frequency	Percentage (%)
Gender	woman	158	45.8
	man	187	54.2
Age	50~54 Years old	34	7.0
	55~60 Years old	172	49.8
	61~64 Years old	112	32.5
	65~70 Years old	28	8.1
	Over 71 years old	9	2.6
Education	primary school and below	16	4.6
	junior middle school	26	7.5
	senior middle school	102	29.6
	undergraduate college	177	51.4
	Master	15	4.3
	doctor	9	2.6
Marital status	unmarried	7	2.0
	married	331	95.9
	dissociation	4	1.2
	bereft of one's spouse	3	0.9
Family annual income	less than 50,000 yuan	15	4.3
	500,000–100,000 yuan	72	20.9
	100,000–200,000 yuan	168	48.7
	more than 200,000 yuan	90	26.1
Working status	full-time job	159	46.1
	hold two or more posts concurrently	15	4.3
	retire	153	44.4
	lose one's job	3	0.9
	other	15	4.3
Whether the work (major) is information technology related	very relevant	52	15.0
	relevant	93	27.0
	uncorrelated	200	58.0
Net age	less than a year	2	0.6
	1–3 Years	25	7.2
	3–5 Years	54	15.7
	more than 5 years	264	76.5
Internet use frequency	nearly everyday	318	92.2
	at least once a week	24	7.0
	at least once a month	2	0.6
	less than once a month	1	0.2
Physical status	healthy	277	80.2
	in general	64	18.6
	poor	4	1.2
Whether to live with children	yes	131	38.0
	no	214	62.0
Permanent residence	city	304	88.1
	rural area	41	11.9

mediation structure. The mediation effect was performed by Preacher and Hayes Bootstrapping test method, which provided a 95% confidence interval estimate of

the mediation effects. According to the data in Table 5, the total effect (−0.615, −0.378) and the indirect effect of pathway “TA-> PU-> BI” (−0.309, −0.077) and pathway

Table 2 Questionnaire questions and reliability test results ¹

Variable	Question number	Item	Cronbach's α after deleting the item	Cronbach's α	AVE	CR
SWB	SWB1	I am satisfied with my life.	0.795	0.813	0.524	0.814
	SWB2	So far, I have got the important things I want in my life.	0.735			
	SWB3	In many ways, my life is close to my ideal life.	0.763			
	SWB4	My living conditions are excellent.	0.762			
PU	PU1	Digitizing public services is valuable to me.	0.846	0.847	0.597	0.815
	PU2	Digitizing public services is conducive to better managing my public affairs.	0.781			
	PU3	Digitizing of public services is good for me.	0.786			
TA	TA1	Using digital public services may make me uncomfortable.	0.895	0.899	0.691	0.899
	TA2	Using digital public services worries me.	0.878			
	TA3	Using digital public services would make me very nervous.	0.861			
	TA4	Using digital public services may be disturbing and confusing to me.	0.840			
SE	SE1	I am confident in using digital public services, even if no one around tells me how to do it.	0.760	0.825	0.613	0.826
	SE2	I have the necessary skills to use digital public services.	0.777			
	SE3	I have the knowledge of using the digital public services.	0.718			
BI	BI1	I plan to use digital public services regularly.	0.787	0.845	0.646	0.845
	BI2	I will always try to use digital public services in my daily life.	0.777			
	BI3	I will use the digital public service in the future.	0.788			
SPA	SPA1	I am just as happy now as when I was young.	0.776	0.801	0.458	0.808
	SPA2	I am just as energetic as last year.	0.761			
	SPA3	The state gets worse as you age.	0.740			
	SPA4	As I get older, I get better (better, worse or the same), as I think.	0.787			
	SPA5	I get less used as I get older.	0.751			

Table 3 Model Fit Indices ³

Model Fit Indices	χ^2/df	RMSEA	CFI	TLI	IFI
result value	2.48	0.066	0.935	0.924	0.935
Standard	< 5	< 0.1	> 0.9	> 0.9	> 0.9

“TA-> SE-> PU-> BI” (−0.477, −0.172) were significant. ⁵

The effect of pathway “TA-> PU-> BI” represented 37.7% of the total effect, and that of pathway “TA-> SE-> PU-> BI” represented 59.3% of the total effect. Direct effect and path “TA-> SE-> BI” mediation effect does not exist, ⁴

Table 4 Results of hypothesis test ⁶

Hypothesis	Pathway	β	S.E.	C.R.	P	Result
H1	SWB<---SPA	−0.737	0.269	−9.785	***	support
H2	TA<---SPA	0.226	0.457	2.612	0.009	support
H3	TA<---SWB	−0.535	0.133	−5.953	***	support
H4	SE<---TA	−0.736	0.036	−13.064	***	support
H5	BI<---TA	0.048	0.043	0.745	0.456	nonsupport
H6	PU<---TA	−0.297	0.034	−4.060	***	support
H7	PU<---SE	0.634	0.063	7.294	***	support
H8	BI<---SE	0.096	0.106	0.935	0.350	nonsupport
H9	BI<---PU	0.961	0.192	7.114	***	support

1.9 Mediation effect analysis ⁸

***: $p < 0.001$ ⁹

Table 5 Results of mediation analysis ¹

Pathway	Effect value	Lower	Upper	P	Proportion
TA-> PU-> BI	-0.189	-0.309	-0.077	0.036	0.377
TA-> SE-> BI	-0.047	-0.147	0.089	0.464	0.093
TA-> SE-> PU-> BI	-0.296	-0.477	-0.172	0.009	0.593
TA-> BI	0.032	-0.067	0.122	0.623	-0.064
Gross effect	-0.500	-0.615	-0.378	0.012	

indicating that technology anxiety and self-efficacy did not directly affect the older adults' digital public service use intention, but technology anxiety and self-efficacy would affect perceived usefulness and affect the use intention of digital public services. Perceived usefulness was the most important antecedent of the use intention of digital public services in the older adults.

Discussion ⁴

This paper examined the structural relationships between self-perception of aging, subjective well-being, technology anxiety, self-efficacy, perceived usefulness and behavior intention in older adults in the context of digital public services. Firstly, the results showed that the self-perception of aging and subjective well-being of older adults affect their technology anxiety, which was consistent with a study finding that psychological factors such as aging self-perception could influence technology anxiety outcomes in older adults [45]. Secondly, technology anxiety indirectly affected the behavior intention to use digital public services, technology anxiety affected the behavior intention to use through the perceived usefulness path and the fully mediated path of self-efficacy and perceived usefulness, but did not directly affect the behavioral intention. It was consistent with a study finding that there was no negative relationship between technology anxiety and behavior intention to use mHealth among older adults in Islamabad, Pakistan [21]. In odds with Nimrod's research, he based on an online survey of 537 Internet users aged 60 and older and found varying levels of technology fear among users, with significant associations between technology fear and Internet usage patterns, including the type and complexity of use. Technical-phobia was also associated with user education, perceived health and well-being [34]. Related to these arguments, our model proposes the underlying mechanisms that technology anxiety might have effect on the behavior intention to use digital public services in older adults.

Negative self-perception of aging might lead to increased technology anxiety. If individuals believed that they had become incompetent or incompetent due to their age, they might experience the anticipation

of difficulties or failure when using unfamiliar digital devices or services, thus increasing anxiety [3, 40, 52], and might lead to fear of self-incompetence, about not being able to master or use technology effectively [25, 51]. The relationship between negative self-perception of aging and technological anxiety might form a vicious circle, and anxiety might hinder the intention to interact with technology, leading to restrictions in the use of experience and contact, which in turn enhanced negative self-perception of their inability to adapt or learn new technologies.

The technology anxiety of the older adults did not directly affect their intention to use digital public services. We concluded that the technology anxiety experienced by older people did not have a direct impact on their willingness to use digital public services, meaning that the discomfort or uneasiness that older people might feel about using technology did not necessarily translate into their unwillingness to use digital public services. Especially in the older population, this view was consistent with the multifaceted view that the factors influenced the adoption of the technology, such as lack of experience of use [34], interest, and perceived usefulness [10], worry about the potential risks of use and the safety and reliability [33]. While technology anxiety might be a concern for some older adults, whose overall intention to use digital public services might also be influenced by other factors. It was also the same conclusion as our study that technology anxiety affected behavior intention to use by influencing perceived usefulness. Because digital public services had become an essential part of daily life, and the service was efficient and transparent [2], which could make the older adults willing to actively use digital public services.

To better understand the mechanism of action between technological anxiety in older adults and their behavior intention to use digital public services, combined with the model, we focused on the role of self-efficacy and perceived usefulness. The relationship between technological anxiety in older adults and their intention to use their technology could be influenced by the complete mediation of perceived usefulness and self-efficacy with perceived usefulness. Analytical validation showed that the direct effect of self-efficacy on behavioral intention was not significant, nor was the technology anxiety to behavioral intention pathway of self-efficacy mediation. Thus, we could conclude that perceived usefulness was a decisive factor in the behavior intention of older adults to use digital public services. Wang confirmed that most older adults had high expectations of usefulness, and perceived usefulness was an important factor influencing their intention to behave [53]. Dequanter demonstrated that perceived usefulness is the most important predictor

of technology acceptance among older adults being cognitively impaired, but also specifically related to satisfaction with social and emotional needs in this group [10]. The conclusions of these studies also confirmed that perceived usefulness was the most important factor affecting the behavior intention to use digital services in older adults.

Identifying and coping with negative aging self-perception in older adults is critical to reducing technology anxiety. In the process of promoting the digital integration of the older adults, we should not only solve the problem of technical skills, but also pay attention to psychological factors, including their attitude and perception of technology. Government, enterprises, communities, families, etc. should formulate relevant measures, on the one hand, improve the information literacy of the older adults, improve the self-efficacy of technology, guide the older adults to establish a positive attitude in the face of information technology, reduce the negative self-aging cognition; on the other hand, improve the subjective happiness, reduce the anxiety of technology, improve the usefulness of technology products, so as to improve their behavior intention to use digital technology. Understanding the interplay between technological anxiety and perceived usefulness is essential to promoting and achieving digital inclusion in older adults.

Conclusion and limitation 3

This study investigates the complex interplay of self-perception of aging, technology anxiety, self-efficacy, subjective well-being, perceived usefulness, and behavioral intention to use digital public services among older adults. By integrating the Technology Acceptance Model (TAM), the aging theory of self-perception, and self-efficacy theory, this study provides valuable insights into how negative self-perceptions of aging and technology anxiety can influence older adults' adoption of digital technologies. Specifically, the results suggest that individuals' self-perception of aging and subjective happiness are critical antecedents of technology anxiety, which in turn affects their technology adoption behaviors. The study contributes to both theoretical and practical understanding by highlighting the psychological factors that shape older adults' willingness to engage with digital public services. The findings hold important implications for policymakers, technology developers, and service providers who aim to improve the digital inclusion of older adults. Addressing technology anxiety, enhancing self-efficacy, and promoting positive aging self-perception could be key strategies for increasing older adults' use of digital services. By applying psychological, management, and information theories, this research adds to the literature on digital public service adoption, offering

a comprehensive framework for understanding how older adults engage with digital technologies. However, several limitations should be considered when interpreting these findings. First, the study's sampling strategy mainly rely on online surveys posted on various platforms. While this approach allowed for a broad reach, it may have introduced a bias by excluding older adults with limited internet access or low digital literacy. Consequently, the sample may not fully represent the diversity of older adults' experiences with technology adoption. Future studies should incorporate a more varied sampling method, such as face-to-face surveys or phone interviews, to include older adults with limited online engagement.

Moreover, the psychological factors, such as self-perception of aging, self-efficacy, and subjective well-being, are inherently complex and interrelated. While the study highlights their role in shaping technology adoption, it is possible that other cognitive, emotional, or social factors, such as privacy concerns or digital trust, also influence older adults' intention to use digital public services. These additional variables need further investigation in future research. Moreover, this study does not account for potential moderating effects that could influence the relationships between the key variables. For instance, demographic characteristics such as age, education, income, or prior technology experience may modify the impact of self-perception or technology anxiety on adoption intentions. Future studies should explore these moderating effects to provide a more nuanced understanding of the factors that facilitate or hinder digital adoption among older adults. Lastly, while this research offers valuable insights into older adults' intentions to use digital public services, it does not consider the longitudinal effects of these factors over time. Technology adoption is a dynamic process, and future research could investigate how the effects of self-perception, anxiety, and self-efficacy evolve as older adults continue to interact with digital technologies. A longitudinal approach would allow researchers to assess whether early interventions to address negative self-perception or technology anxiety lead to sustained changes in technology usage behavior.

To address these limitations, future studies could consider the following perspectives. Firstly, future studies may consider adopting a diversified sampling method and enlarging the sample size. Future research could employ a more varied sampling strategy that combines both online and offline channels. For example, while continuing to recruit participants through online platforms, researchers could also conduct field surveys or interviews in offline settings such as community centers, senior activity clubs, and nursing

homes. This approach would ensure that the sample comprehensively covers older adults from different types of communities, regions, and varying levels of internet access. By doing so, the representativeness of the sample would be enhanced, making the research findings more generalizable and contributing to the development of a more robust theoretical model for the adoption of digital public services among older adults. Secondly, future research could incorporate additional psychological determinants, such as digital trust, privacy concerns, or attitudes toward technology, to provide a more comprehensive model for the adoption of digital public services. Furthermore, examining the moderating effects of demographic variables, such as age, education level, and prior experience with digital technologies, will help to gain a deeper understanding of how these factors influence older adults' adoption of digital public services. Lastly, conducting longitudinal studies to track changes in older adults' perceptions and behaviors over time will help evaluate the long-term effects of interventions aimed at reducing technology anxiety or enhancing self-efficacy.

Supplementary Information 2

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Supplementary Material 1. 4

Authors' contributions 5

JA designed the study and revised this manuscript; XYZ wrote this manuscript and conducted the data analysis; XYZ, KXW, and ZYX revised this manuscript; WH provided resources and funding; ZS and JLA collected data.

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Data availability 9

The data sets used and analyzed in this study are available from the corresponding author on reasonable request.

Declarations 11

Ethics approval and consent to participate 12

This study was conducted in accordance with the Declaration of Helsinki and approved by the ethics committee of First People's Hospital of Changshu City. Informed consent was obtained from all participants.

Consent for publication 14

Not applicable. 15

Competing interests 16

The authors declare no competing interests. 17

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